International Marketing Analytics

Experiments and A/B testing

Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business

arne.floh@wu.ac.at
Office hours: book online (Google Calendar)

WT 2026/27

Today

Why experiments → A/B test → 2 × 2 designs → Python → In class

Theory, Parts 1 to 3

- Why only a random split shows cause and effect
- Design and read an A/B test
- Factorial designs and interactions

Practice, Parts 4 and 5

- polars, scipy, statsmodels, plotnine
- Online quiz and its answers
- Coding exercise: the operator's e-mails
- The same tasks with Posit Assistant

Question of the day: which e-mail makes more customers click, and how do we know it was the e-mail?

Before we start: did the ad triple sales?

Customers who saw a retailer's Facebook ad bought about four times as often as those who did not (+316 %).

Who decided who saw the ad?
Was it a coin flip?

Why might viewers buy more anyway?
Name one reason that is not the ad.

What should we compare instead?
Think of two groups that are alike.

Source: Gordon et al. (2019), Marketing Science, Study 4

Our example: a retention e-mail in two markets

The operator: the mobile operator from the churn session e-mails customers in Germany and Spain about its loyalty offers.

The question: which subject line makes more customers click?

A

Inbox

Loyalty team

Your loyalty offers are here

B

Inbox

Loyalty team

Anna, your loyalty offers are here

A = generic (control)

B = with the first name (variant)

Two tests: A versus B on 40,000 customers, then a 2 × 2 test of first name × discount on 40,000 more.

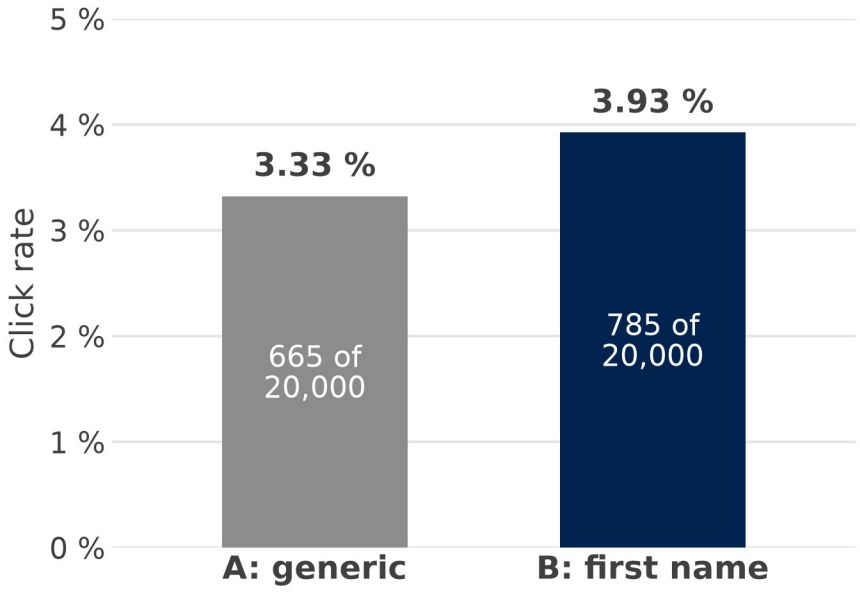
Simulated data (email_ab.csv, email_factorial.csv), built for this course

The data: one row per customer

Column	Meaning	email_ab.csv	email_factorial.csv
customer_id	customer number	8	100044
country	DE or ES	DE	DE
device	mobile or desktop: where the e-mail was read	mobile	mobile
version	A = generic, B = first name; A/B test only	A	–
personal	1 = first name in the subject line; 2 × 2 only	–	1
discount	1 = 10 % off the next bill; 2 × 2 only	–	1
clicked	1 = clicked the link: the outcome	1	1
unsubscribed	1 = left the mailing list	0	0

Files: `data/abtest/email_ab.csv`, `email_factorial.csv` simulated, 40,000 customers each

First look: click rates of e-mail A and B



+0.60 pp

percentage points:
3.93 % − 3.33 %

+18 %

relative lift:
0.60 / 3.33

Two questions for today: did the name cause the gap, and is it more than chance?

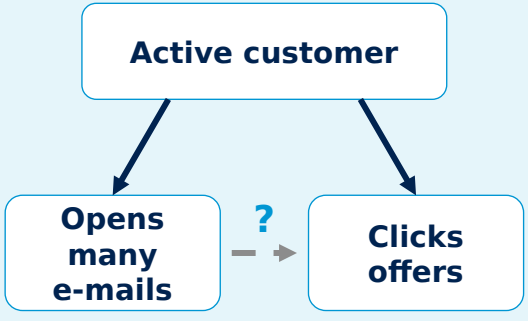
Simulated data (email_ab.csv)

Why experiments

Part 1 of 5 · cause, counterfactual, coin flip

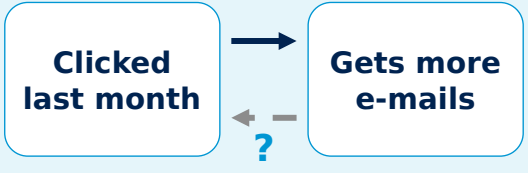
Correlation is not causation: three traps

Confounder



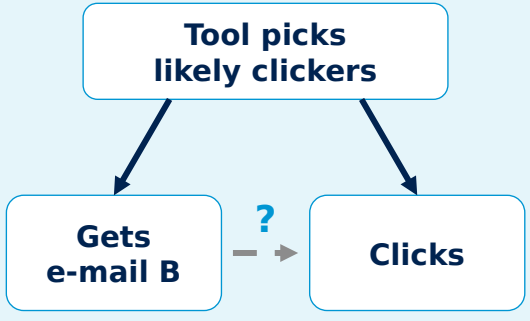
Activity drives both.

Reverse causality



Clicks drive the mailing, not the other way round.

Selection



The algorithm, not the e-mail, made B look good.

The fix: a coin flip decides who gets which e-mail, so nothing else can drive both.

The counterfactual: two worlds, one customer



World B: e-mail B (her name)
Does she click?

Effect for Anna =
World B − World A

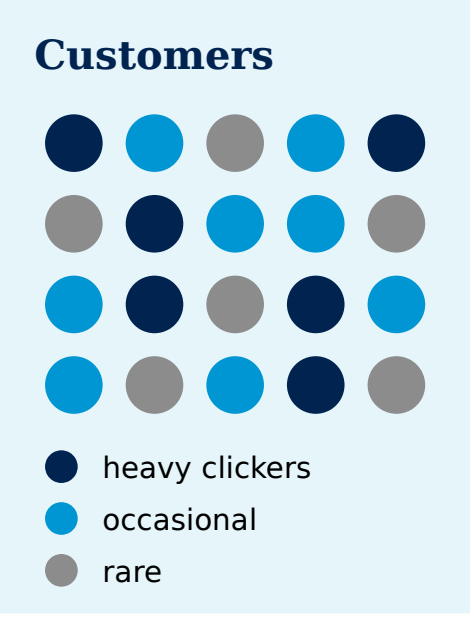
World A: e-mail A (generic)
Does she click?

Problem: we only ever see one world for Anna.

Solution: a random group A stands in for the world Anna did not live.

Average effect = click rate B − click rate A: +0.60 pp.

A coin flip makes the groups alike



A **Group A: generic**

B **Group B: first name**

In our data	Group A	Group B
Customers	20,000	20,000
Read on mobile	59.5 %	59.9 %
In Spain	50.0 %	50.0 %

Random assignment: the groups differ only by chance and by the e-mail.

Simulated data (email_ab.csv); dots: illustration

What is an A/B test?

Customers → Random split → **A or B** → Measure clicks → Decide

A = control: the current e-mail

B = variant: the same e-mail with one change, the treatment

A/B/n

several versions of one part

Factorial

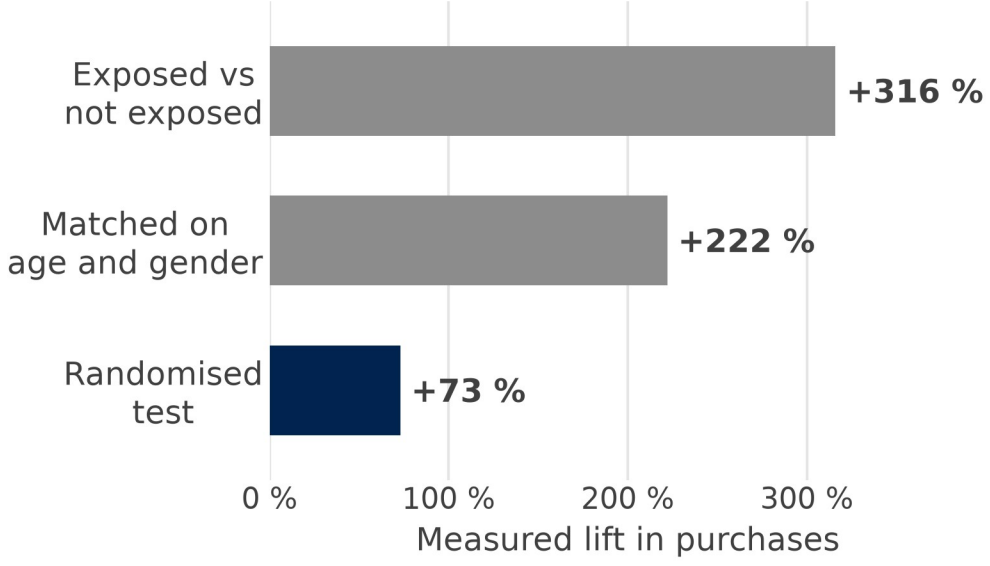
two changes at once

A/A

the same e-mail twice

Effect = click rate B − click rate A, measured on the same day, with the same kind of customers.

The warm-up resolved: naive vs randomised



4× the naive comparison overstates the effect: **316 vs 73**

Why? The platform showed the ad to people who were more likely to buy anyway.

The ad finds buyers more than it creates them: only a random control group shows the effect.

Source: Gordon et al. (2019), Marketing Science, Study 4

Designing and analysing an A/B test

Part 2 of 5 · variables, power, checks

What we change, what we measure

Independent variable

The treatment: the subject line

A generic | **B** first name

causes?

Dependent variable

The outcome: clicked (1/0), summarised as the click rate

1 clicked | **0** no click

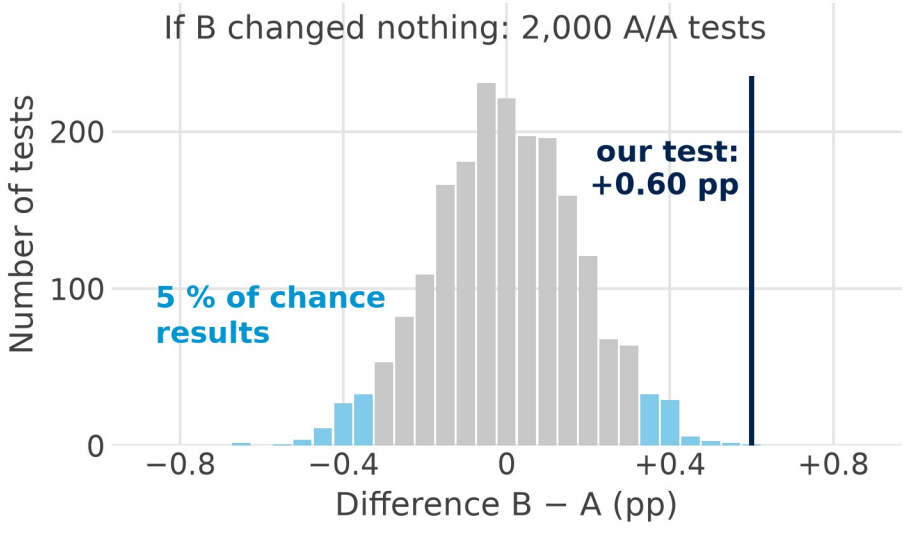
Key metric: the click rate decides – fixed before the e-mail is sent

Guardrail: the unsubscribe rate must not get worse

Held constant: body text, send day, offer page

One change, one key metric: decide both before the start.

Is +0.60 pp more than chance?



Guardrail: unsubscribes 0.31 % (A) vs 0.41 % (B), p 0.079

+0.60 pp — **B − A:** 3.93 % − 3.33 %

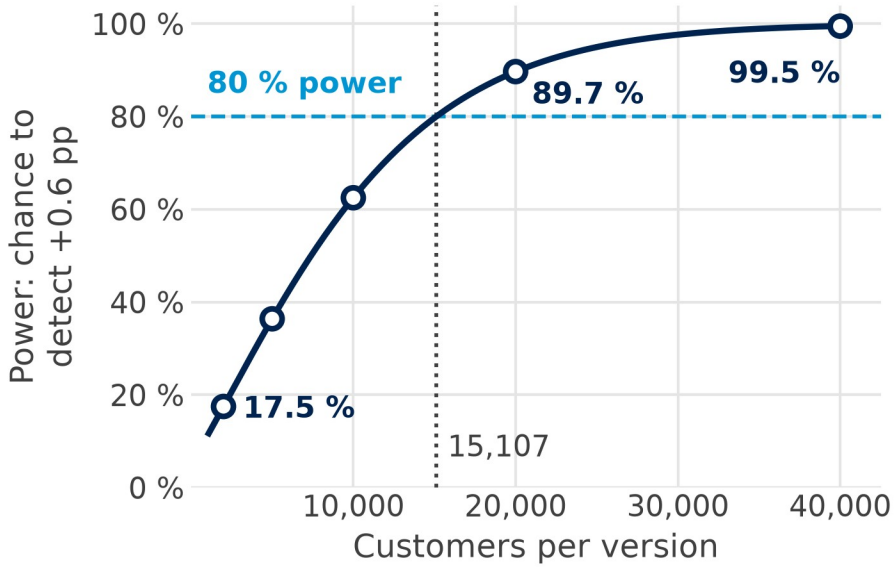
0.0013 — **p-value:** how often chance alone gives a gap at least this large

0.23–0.97 — **95 % CI** in pp: the effect is plausibly between these

Report the effect in pp with its CI, not only the p-value.

Simulated data (email_ab.csv); 2,000 simulated A/A tests

How many customers? Sample size and power



15,107 — **per version** for +0.6 pp (3.3 → 3.9 %), 80 % power

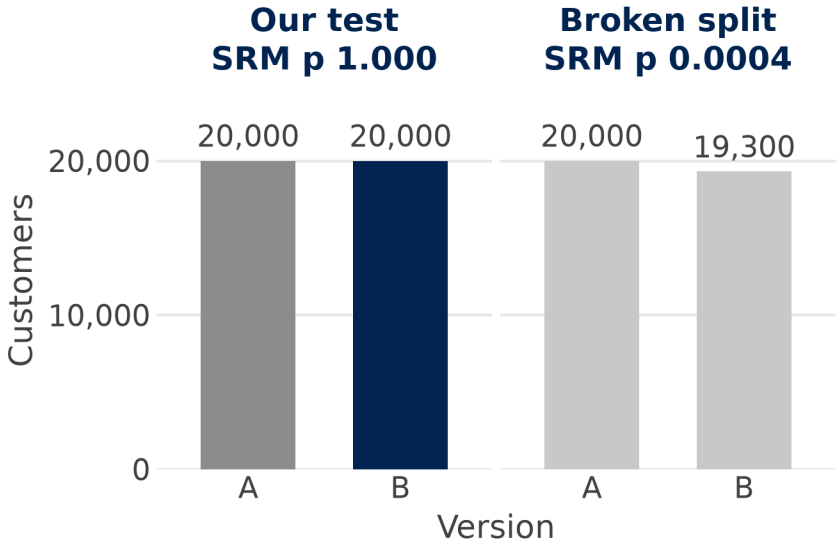
58,072 — **for +0.3 pp:** half the effect, four times the customers

5,715 — **for +1.0 pp**: a large effect needs few

Plan the sample before you send, and do not stop early when p dips below 0.05.

Simulated data (email_ab.csv); power for 3.3 → 3.9 %, 5 % level

Check the set-up: the split and an A/A test



Sample ratio mismatch (SRM)

Planned 50:50, got something else by more than chance? Stop: find the cause before reading any result.

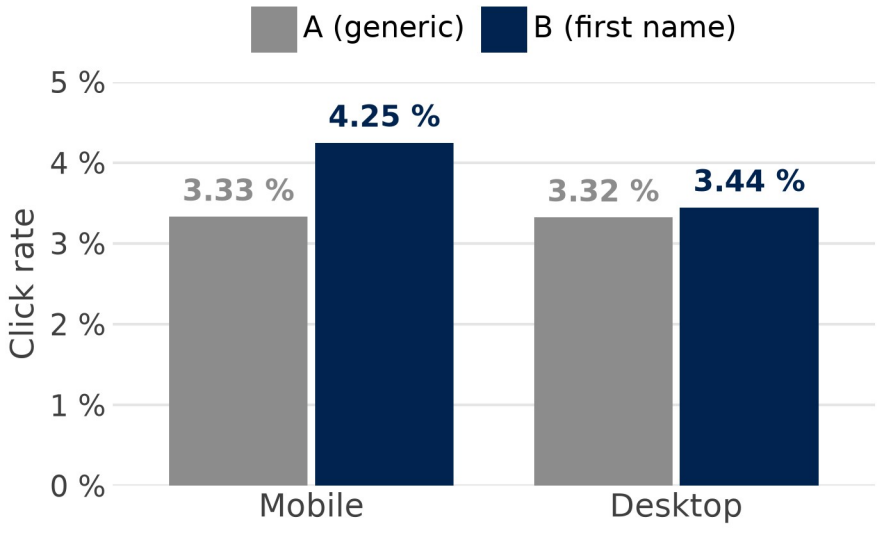
A/A test

The same e-mail to both groups: about 5 % come out “significant” (5.9 % in our 2,000 runs).

Check first: if the split test gives p below 0.001, fix the set-up before you read any result.

Simulated data (email_ab.csv); the broken split is invented

Does B work for everyone? Mobile vs desktop



Labels rounded; gaps from unrounded values

+0.92 pp **on mobile** (23,885 customers)

+0.13 pp **on desktop** (16,115 customers)

+0.79 pp **Interaction:** the effect of B depends on the device (p 0.038)

A segment found after the test is a hypothesis: test it again (Part 3).

Simulated data (email_ab.csv)

Can we trust it? Internal and external validity

Internal validity

- **Did the e-mail cause it here?** random split, same day, same kind of customers
- **Threat:** a broken split, or a platform that shows A and B to different people
- **Check:** the split, and who actually got A and B

External validity

- **Will it hold elsewhere?** other markets, seasons, channels, customers
- **Threat:** a winner here may fail elsewhere
- **Check:** repeat before roll-out (slide 27)



A vs B = ad **plus** audience

Randomisation buys internal validity; external validity needs a repeat.

Source: Braun & Schwartz (2025), Journal of Marketing

Factorial designs and interactions

Part 3 of 5 · factors and interactions

From A/B to 2 × 2: two factors, four cells

Factor 1: first name	Factor 2: discount — No discount	Factor 2: discount — 10 % discount
Generic	**Inbox** 10,000 customers Loyalty team **control** **Your loyalty offers are here**	**Inbox** 10,000 customers Loyalty team **Your loyalty offers are here** **10 % off your next bill**
First name	**Inbox** 10,000 customers Loyalty team **Anna, your loyalty offers are here**	**Inbox** 10,000 customers Loyalty team **Anna, your loyalty offers are here** **10 % off your next bill**

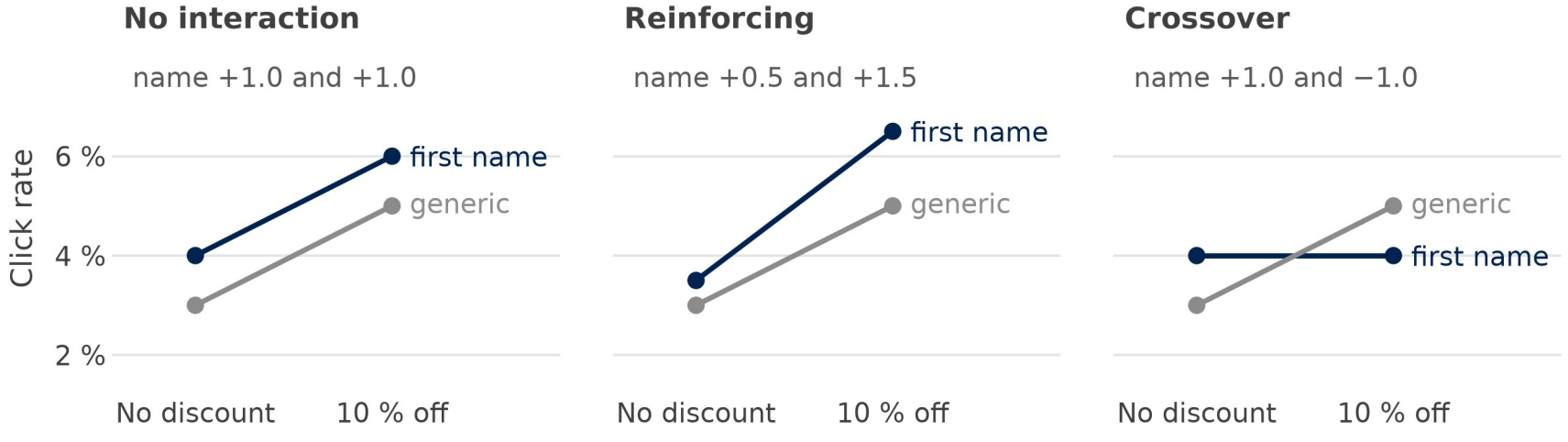
Factor: an independent variable we change (here: first name, discount)

Level: a version of a factor (no / yes)

Cell: one combination: 2 × 2 = 4 e-mails

Randomised: each customer gets one of the four e-mails at random.

Interaction: a difference in differences



Interaction = simple effect of the name with discount − without (the gap between the lines): **0, +1.0, −2.0 pp**

Parallel lines: no interaction. Fan out or cross: the effect of one factor depends on the other.

Illustration: invented round numbers

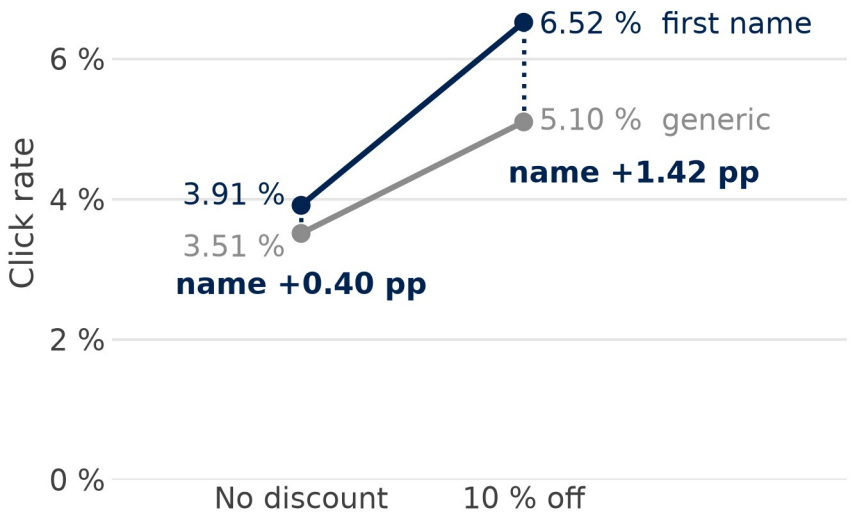
Our 2 × 2 result: the name works with the discount

	No discount	10 % discount
Generic	3.51 %	5.10 %
First name	3.91 %	6.52 %

Click rate per cell, 10,000 customers per cell

Name alone
+0.40 pp

Name + discount
+1.42 pp



The name adds 0.40 pp alone and 1.42 pp with the discount: an interaction of +1.02 pp.

Simulated data (email_factorial.csv)

By hand: simple effects, main effects, interaction

Effect of the first name	Calculation	pp
Simple effect, no discount	3.91 − 3.51	+0.40
Simple effect, with discount	6.52 − 5.10	+1.42
Main effect (average)	(0.40 + 1.42) / 2	+0.91
Interaction (difference)	**1.42 − 0.40**	**+1.02**

Rule

```
simple effect
  = within one level
main effect
  = average of the
    simple effects
interaction
  = difference of the
    simple effects
```

Discount, the same way: +1.59 and +2.61 pp, main effect +2.10 pp, interaction +1.02 pp

With an interaction, report the simple effects: the average (+0.91 pp) fits neither e-mail.

Simulated data (email_factorial.csv)

Regression and ANOVA: one model, two read-outs

`smf.ols("clicked ~ personal * discount", data=d)` coefficients × 100 = pp

Term	Coef. (pp)	p	Reads as
Intercept	3.51 %		generic, no discount
personal	+0.40	0.183	name, without discount
discount	+1.59	< 0.001	discount, generic e-mail
personal:discount	**+1.02**	**0.017**	**the interaction**

Two-way ANOVA	F
personal	18.32
discount	97.56
personal:discount	**5.75**

Interaction: p 0.017, the same test

'personal' is not the main effect: with a product term it is the name's effect without discount (n.s.); the ANOVA row tests the main effect.

Same data, same interaction test: +1.02 pp, p 0.017.

Simulated data (email_factorial.csv); OLS on 0/1 clicks (linear probability model)

The logit reads it differently: use pp

Same four cells	Name alone	Name with discount	Interaction	p
Click rate, pp (OLS)	+0.40	+1.42	+1.02	0.017
Log-odds (logit)	+0.11	+0.26	+0.15	0.124

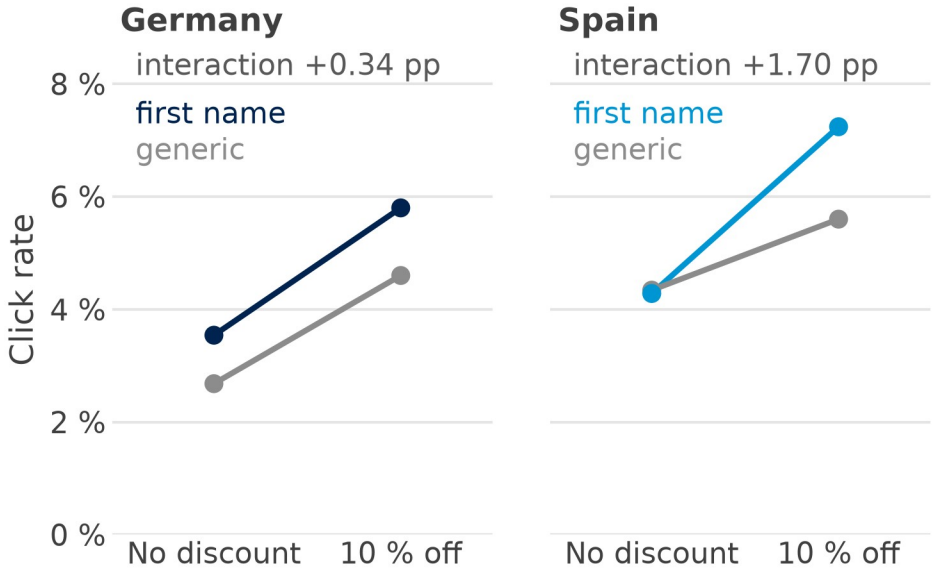
Why they differ: the discount raises the base rate, and the same odds lift adds more pp at 5.10 % than at 3.51 %. The logit removes that part, so less interaction is left (+0.15 in log-odds).

Two questions: the logit asks whether the odds ratio changes; the manager asks how many more customers click.

Read interactions in percentage points, as in the logit deck (slide 21).

Simulated data (email_factorial.csv)

The market as a moderator: Germany vs Spain



+0.34 pp **Germany** (p 0.546)

+1.70 pp **Spain** (p 0.008)

p 0.110 **Spain − Germany** (+1.36 pp): within chance

'Significant in Spain, not in Germany' is not evidence of a difference: test the moderator, then repeat per market.

Simulated data (email_factorial.csv)

Design rules for factorial tests

Two separate A/B tests · 40,000 customers

10,000	10,000	**Name:** 10,000 vs 10,000
10,000	10,000	**Discount:** 10,000 vs 10,000

One 2 × 2 · 40,000 customers

10,000	10,000	**Name:** 20,000 vs 20,000
10,000	10,000	**Discount:** 20,000 vs 20,000

+ the interaction: only the factorial can test it

1. **Twice the use of each customer:** every main effect compares 20,000 with 20,000.
2. **Interactions need more:** about four times the customers of a main effect of the same size (SE 0.43 vs 0.21 pp).
3. **Keep it small:** 2 × 2 or 2 × 3; a factorial with 4 × 6 = 24 cells (in industry: a multivariate test) spreads customers thin.

Plan the interaction in advance: which one, how big, how many customers.

Illustration

The analyses in Python

Part 4 of 5 · polars, scipy, statsmodels, plotnine

The packages and functions

Task	Package	Function
Load and count	polars	`pl.read_csv(); group_by().agg()`
Check the split (SRM)	scipy	`stats.chisquare(counts)`
z-test and CI	statsmodels	`proportions_ztest(); confint_proportions_2indep()`
Sample size and power	statsmodels	`proportion_effectsize(); NormalIndPower().solve_power()`
Factorial regression	statsmodels	`smf.ols("clicked ~ personal * discount")`
Two-way ANOVA	statsmodels	`anova_lm(m, typ=2)`
Tables and plots	great_tables, plotnine	`GT(); ggplot, geom_line`

Step 1: load and check the split

Python: polars, scipy

```
import polars as pl
from scipy import stats

ab = pl.read_csv("../../data/abtest/email_ab.csv")

split = (ab.group_by("version")
         .agg(pl.len().alias("n"),
              pl.col("clicked").sum().alias("clicks"),
              pl.col("clicked").mean().alias("click_rate"))
         .sort("version"))
stats.chisquare(split["n"].to_list()).pvalue  # SRM
split
```

Version	n	Clicks	Rate
A	20,000	665	3.33 %
B	20,000	785	3.93 %
SRM p			1.000

Customers and clicks per version

Check the split first: an SRM p below 0.001 means stop and fix the set-up.

Step 2: difference, z-test, CI and power

Python: statsmodels

```
from statsmodels.stats.proportion import (
    proportions_ztest, confint_proportions_2indep,
    proportion_effectsize)
from statsmodels.stats.power import NormalIndPower

x, n = split["clicks"].to_list(), split["n"].to_list()
z, p = proportions_ztest([x[1], x[0]], [n[1], n[0]])
lo, hi = confint_proportions_2indep(x[1], n[1],
                                    x[0], n[0])   # B - A

es = proportion_effectsize(0.039, 0.033)
NormalIndPower().solve_power(es, alpha=0.05, power=0.8)
```

Result	Value
Difference B − A	+0.60 pp
z	3.21
p	0.0013
95 % CI	[0.23, 0.97] pp
n for +0.6 pp	15,107

n: customers per version for 80 % power

Read it as: B adds about 0.6 pp; the sample was planned for this size.

Step 3: the factorial regression and plot

Python: statsmodels, plotnine

```
import statsmodels.formula.api as smf
from plotnine import aes, geom_line, geom_point, ggplot

fa = pl.read_csv("../../data/abtest/email_factorial.csv")
m = smf.ols("clicked ~ personal * discount",
            data=fa.to_pandas()).fit()
m.params * 100                      # in pp

cells = (fa.group_by("personal", "discount")
         .agg(pl.col("clicked").mean().alias("rate")))
(ggplot(cells, aes("discount", "rate",
                   colour="factor(personal)"))
 + geom_line() + geom_point())
```

Term	pp	p
Intercept	3.51 %	–
personal	+0.40	0.183
discount	+1.59	< 0.001
personal:discount	**+1.02**	**0.017**

Each coefficient is a cell difference (slide 25)

Read the product term in pp: +1.02, p 0.017; the plot shows it as lines that fan out.

In-class exercise

Part 5 of 5 · quiz, exercise, Posit Assistant

Online quiz: experiments in five questions

1 Why can we read B − A as the effect of the first name?
A both groups are large B the coin flip made A and B alike C clicks have no noise

2 Click rate 3.0 % with A, 3.6 % with B. The difference is …
A +0.6 pp, or +20 % relative B +0.6 % C +20 pp

3 15,107 customers per version detect +0.6 pp. For +0.3 pp you need about …
A 7,500 B 30,000 C 60,000

4 Planned 50:50, got 20,000 in A and 19,300 in B (SRM p < 0.001). You …
A report the result B stop and look for the bug C drop 700 customers from A

5 2 × 2: the name adds +0.4 pp without, +1.4 pp with the discount. The interaction is …
A +1.0 pp B +0.9 pp C +1.8 pp

3 minutes alone, then compare with your neighbour.

Answers: in class

Your turn: test the e-mails in Python

Pairs, with packages: 35 minutes	Plenary, answers: 5 minutes	Posit Assistant: 10 minutes

1. **Check:** the split and the share on mobile per version, in each market (Step 1)
2. **Test:** B − A in Germany and in Spain, in pp with 95 % CI; does it differ? (Steps 2, 5)
3. **Plan:** customers per version for 3.3 % → 3.8 % with 80 % power (Step 2)
4. **2 × 2:** cell table, interaction plot, regression; which e-mail should be sent? (Steps 3, 6)

Deliverable: one slide per pair with the interaction plot and two sentences for the CRM manager.

File: `sessions/01-foundations/ab_exercise.qmd` (code to reuse: ab_testing.qmd)

Then: the same tasks with Posit Assistant

How: code first, then open Posit Assistant, type @ab_exercise.qmd, one task per prompt

Prompt for task 1

```
@ab_exercise.qmd Task 1. Check the split of
email_ab.csv in each country: counts,
SRM p-value and the share on mobile
per version.
```

Prompt for task 2

```
Task 2. Estimate B – A in Germany and
in Spain in pp with a 95 % CI. Do the
markets differ, and how would you
test that?
```

Prompt for task 3

```
Task 3. How many customers per version
detect 3.3 % → 3.8 % with 80 % power?
Explain power in one sentence for
a manager.
```

Prompt for task 4

```
Task 4. Fit clicked ~ personal * discount.
Explain each coefficient in pp and
recommend one e-mail; check the
unsubscribes too.
```

Check every number against your own code: you own the result.

Open data for A/B labs

Hillstrom e-mail

64,000 customers of a retailer

Design: 3 random groups: men's e-mail, women's e-mail, none

Outcome: visits, purchases, spend

Access: free CSV, no licence stated

Upworthy

4,873 headline tests of a news site

Design: random within each test

Outcome: clicks

Licence: CC BY 4.0 (OSF)

Criteo uplift

14 million rows, version 2.1

Design: random treatment

Channel: online ads

Licence: CC BY-NC-SA

The licence decides: commit only data you may share; link the rest.

Links and loading code: `ab_testing.qmd` (appendix)

Sources: dataset checks, Oct 2026

Key takeaways

1. **Seeing is not causing:** compare random groups, not those who got it with the rest
2. **A coin flip makes A and B alike:** B − A is the effect; report it in pp with its CI
3. **Plan and check:** the sample size before the send, the split before the result
4. **Factorial designs test two changes at once:** with an interaction, report the simple effects
5. **Read interactions in pp, test moderators:** “significant here, not there” is not a difference

Next: run sessions/01-foundations/ab_testing.qmd; it reproduces every number of the deck.

References

Experiments and methods

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Code and data: `sessions/01-foundations/ab_testing.qmd`

Thank you. Questions and discussion

VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

INSTITUTE FOR INTERNATIONAL BUSINESS
Welthandelsplatz 1, 1020 Vienna, Austria

DR ARNE FLOH
Senior Lecturer in Marketing

F +43-31336-6367
arne.floh@wu.ac.at
www.wu.ac.at/en/welthandel/arne-floh
Office hours: book online